

Exact Lower Bound for Equitable Harmonic Hosting Capacity

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Abstract—This paper introduces an optimization method to determine the lower bound on the harmonic hosting capacity in power systems. The novelty of this work lies in its comprehensive approach, which includes detailed network and unit representations, individual and system harmonic limits, and a variety of fairness criteria for equitable distribution of the available harmonic budget. The proposed method employs both non-linear and second-order cone models to analyze harmonic power flows, ensuring accurate and tractable solutions. Numerical illustrations show that the second-order cone relaxation approach, determines the harmonic hosting capacity lower bound in an exact and tractable way. The practical applicability of the approach is demonstrated on the Rte 1888 bus system, where the effects of the different fairness criteria and system wide harmonic limits are discussed. The optimization model is made available as an open-source software package for further research and practical application in power system planning.

Index Terms—Harmonic distortion, fairness, hosting capacity, mathematical optimization.

I. INTRODUCTION

A. Background and Motivation

REACHING a net zero economy by 2050 requires a massive undertaking in electrification of the energy system. Over the last two decades, renewable power generation, i.e., wind and photovoltaic generation, has successively increased. This trend is expected to accelerate further towards 2050 [1]. Additionally, to achieve the decarbonisation goals, the demand also needs to be decarbonised, which requires the substitution of fossil fuel by electricity for transport, heating and industrial processes. In Europe, various studies show that the total electricity demand will triple by 2050 [2], [3].

Renewable energy sources and new types of electrical loads are almost exclusively power electronic interfaced assets. This brings a number of challenges for the planning and operation of future power grids. Power electronic interfaced assets behave fundamentally different compared to classical electrical machines, or resistive and inductive loads. One of the major concerns is the increase in the harmonic distortion causing higher network losses and resonances as direct consequences.

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Indirect consequences include malfunctioning of protection systems and, over time, the additional heating of transformers and power cables causing accelerated aging. Furthermore, motors, inverters and other devices may be sensitive to the presence of voltage distortion.

Sun et al. highlight that the traditional harmonic studies performed at the initial planning stage may no longer be appropriate [4]. Considering the transition to a heavily power electronic dominated system, a central question to be asked is: “what is the maximum harmonic distortion, e.g., number of large-scale converters, that can be reliably hosted in a given power system?”. In literature, this concept is referred to as the hosting capacity of a power system [5], [6]. The hosting capacity can be determined in terms of a number of different performance limits, e.g., transformer overload, feeder ampacity, etc. Recently, harmonic distortion has gained attention in literature as a relevant performance limit. Nakhodchi et al. have shown that harmonic distortion as a performance limit is often more critical than the transformer rating [7]. There is an important gap in literature for a harmonic hosting capacity model that includes the following features:

- 1) *network representation*, including: (a) nodal or full network representation, (b) non-linear (network) impedance in the frequency domain, (c) phase-(un)balanced representation, and (d) detailed treatment of power transformer.
- 2) *unit representation*, including: (a) generic unit, and (b) equitable distribution of the harmonic hosting capacity.
- 3) *harmonic limits*, including on: total harmonic distortion, individual harmonic distortion, root-mean-square ampacity, power factor, and voltage unbalance.
- 4) *stochastic representation*, including: (a) unit harmonic injection, (b) background harmonic distortion, (c) network parameters, (d) network exploitation, and (e) harmonic limits.

In the context of power system planning, this work aims to address the features in categories 1), 2) and 3) by exactly and tractably determining the lower bound for the harmonic hosting capacity, considering harmonic distortion throughout the entire network under different equitable distributions of the harmonic distortion budget.

B. Literature Review

There are several standards (std.) and, more recently, limited academic literature (lit.) that present an approach to determine the harmonic hosting capacity of a power system. Table I enumerates their mathematical method and features with respect

TABLE I

LITERATURE REVIEW ON HARMONIC HOSTING CAPACITY, ENUMERATING FEATURES WITH RESPECT TO THE CATEGORIES ESTABLISHED IN § I-A. THE LAST LINE OF THE TABLE SHOWS THE FEATURES INCLUDED IN THE PRESENTED PAPER.

ref.	class	method ¹	1) ntw. repr.				2) unit repr.		3) harm. lim.					4) stoch. repr.					tract. ²	
			(a)	(b)	(c)	(d)	(a)	(b)	thd	ihd	rms	pf	unb.	(a)	(b)	(c)	(d)	(e)	# n	t[min]
[8]	std.	s-ae	nodal	×	bal.	×	✓	×	✓	✓	×	×	×	×	×	×	×	×	-	-
[9]	std.	s-ae	nodal	×	bal.	×	✓	×	✓	✓	×	×	×	×	×	×	×	×	-	-
[10]	std.	s-ae	nodal	×	bal.	×	✓	×	✓	✓	×	×	×	×	×	×	×	×	-	-
[11]	lit.	s-ae	nodal	✓	bal.	×	✓	×	×	✓	×	×	×	×	×	×	×	×	14	-
[7]	lit.	mc-ae	nodal	✓	bal.	×	×	×	✓	✓	×	×	×	✓	×	×	×	×	64	-
[12]	lit.	mc-pf	full	✓	unb.	×	×	×	✓	×	✓	×	✓	×	×	×	×	×	8773	320
[13]	lit.	mc-pf	full	✓	bal.	×	×	×	✓	✓	✓	✓	×	✓	✓	×	×	×	3	-
[14]	lit.	nl	full	✓	bal.	×	×	×	✓	✓	×	×	×	✓	×	×	×	×	16	-
[15]	lit.	nl	full	✓	bal.	×	×	×	✓	✓	×	×	×	✓	×	×	×	×	5	-
[4]	lit.	nl	full	✓	bal.	×	×	×	✓	✓	×	×	×	✓	×	×	×	✓	16	30
This work	lit.	soc	full	✓	bal.	✓	✓	✓	✓	✓	✓	×	×	×	×	×	×	×	1880	< 6

to the categories established in § I-A. Furthermore, the last row of Table I shows the features included in the presented paper. The last two columns of Table I shows the tractability of the proposed methods in the literature. It can be seen that no tractable optimal method exists in literature which incorporates all features in categories 1), 2) and 3).

1) *Network representation*: Two possible network representations are considered in different harmonic hosting capacity studies, i.e., nodal or full network representation. A nodal network representation omits the interaction between different sources of harmonic distortion, located at different nodes in the system. For example, the nodal network representation is unable to account for the effects of a series resonance, i.e., harmonic current injection into the considered node may not cause a local violation of the harmonic voltage limits, but causes an external violation somewhere further remote in the system. All but one reference, i.e., [12], considers a phase-balanced network representation, even those studying distribution systems, but neglect to model power transformers in detail [16]. In contrast to the standards which derive the harmonic network impedance on the local short-circuit impedance, recent literature considers the frequency-dependent of the impedance based on those of the individual components, reflecting the resonances in the network.

2) *Unit representation*: In contrast to the standards, most harmonic hosting capacity studies in the literature consider one specific application, e.g., hosting capacity of photovoltaic systems or electric vehicle charging stations, accounting for harmonic limits. However, from the perspective of the power system operator, such studies are limiting as power system operators have no knowledge of the specific (future) assets of its consumers. Furthermore, in the European context, power system operators are regulated and have to treat their customers on a fair and equal basis. Consequently, in this context, the harmonic hosting capacity should be distributed amongst its consumers based on some fairness principle. The application of fairness in the context of harmonic hosting

capacity remains under-investigated in the literature [11].

3) *Harmonic limits*: The main harmonic limits considered in harmonic hosting capacity studies are on total harmonic distortion (thd), individual harmonic distortion (ihd), and root-mean-square value (rms) of the nodal voltage or edge ampacity. Additionally, power factor limits are considered in [13], and, given the phase-unbalanced network representation, voltage unbalance limits are considered in [12].

4) *Stochastic representation*: Different sources of uncertainty may have an impact on the harmonic hosting capacity, including: (a) unit harmonic injection, (b) background harmonic distortion, (c) network parameters, and (d) network exploitation, as they inherently impact the harmonic limits. Uncertainty on the unit harmonic injection and harmonic background are considered in [4], [7], [13]–[15]. However, the impact of uncertainty on network parameters and exploitation remain under-investigated in the literature.

C. Contributions and Outline

In this paper, a holistic one-shot methodology is presented to exactly and tractably determine the lower bound on the steady-state harmonic hosting capacity for an entire phase-balanced three-phase power system in a deterministic context. As the presented method does not consider uncertainty on the angle of the injected harmonic current, rather assumes a constant angle relative to the reference, its result is the lower bound on the steady-state harmonic hosting capacity. The problem is formulated from the perspective of the power system operator, i.e., without knowledge of the (future) assets of its costumers. Consequently, the harmonic hosting capacity is determined in terms of harmonic current magnitude using a fairness principle. Two exact formulations of the problem in the rectangular current-voltage variable space are presented: non-linear and second-order cone, both incorporating:

- exact representation of the network impedance in the frequency domain based on the harmonic impedance of the individual grid components,
- detailed treatment of transformers including the effects of their configuration,
- enforcement of (harmonic) limits given in relevant standards on bus voltage and component ampacity, and
- equitable distribution of the harmonic hosting capacity following a fairness principle.

¹The possible mathematical methods are abbreviated as: (s-ae) single analytical expression, (mc-ae) monte-carlo analytical expression, (mc-pf) monte-carlo power flow, (nl) non-linear optimal power flow, and (soc) second-order cone optimal power flow.

²The tractability of a method is expressed in terms of size of the network studied, i.e., number of nodes #n, and the solve time t given in minutes.

The models and numerical illustrations presented in the paper are made freely available as part of the JULIA-package: HARMONICPOWERMODELS.JL³.

The remainder of the paper is organized as follows: Section II presents the mathematical formulation of the harmonic hosting capacity problem. Section III shows numerical illustration results for a radial section of a real-world industrial power system and for a large-scale transmission system. Section IV concludes the paper.

II. MATHEMATICAL FRAMEWORK

This section presents the mathematical framework for the harmonic hosting capacity problem of a phase-balanced three-phase power system. To this end, first, the impact of incorporating harmonics into a phase-balanced power flow problem is discussed. Second, two formulations, i.e., non-linear and second-order cone, of the harmonic hosting capacity problem are introduced. The harmonic hosting capacity problem determines the maximum amount of harmonic current injected by units, given a fairness principle, without violating voltage and current bounds.

In the following table, a general overview and generic examples of the notation used throughout the mathematical framework are provided.

element	notation	example
index	lower-case italic	x
set	calligraphic	$\mathcal{X}$
variable	sans serif	X
parameter	italic	$\bar{X}$
complex quant.	bar notation	$\bar{X}, \bar{X}$
complex operator		j

A. Harmonic Power Flow

Under steady-state conditions, current and voltage waves are not necessarily purely sinusoidal, but rather are, more generically, periodic waves. The Fourier transform may be used to decompose periodic waves into the superposition of sinusoidal waves, i.e., harmonics. A harmonic voltage or current oscillates at a multiple $h \in \mathcal{H} \subset \mathbb{N}$ of the fundamental frequency, e.g., 50 Hz or 60 Hz⁴. A phase-balanced three-phase power system allows a single-phase representation of the system for all individual harmonic. Specifically, the harmonics are classified into three types:

- 1) positive sequence $h \in \mathcal{H}^+ \subset \mathcal{H}$: $h = 1 + 3n$,
- 2) negative sequence $h \in \mathcal{H}^- \subset \mathcal{H}$: $h = 2 + 3n$, and
- 3) zero sequence $h \in \mathcal{H}^0 \subset \mathcal{H}$: $h = 3 + 3n$,

where $n \in \mathbb{N}_0$. Historically, in such power systems, only odd-order harmonics, i.e., $h \in \{1, 3, 5, \dots\}$ were present, whereas even-order harmonics, i.e., $h \in \{2, 4, 6, \dots\}$ were only present given half-wave or defective full-wave rectifiers. However, with the advent of active front-end converters, all harmonics are expected. Furthermore, in such power systems, zero-sequence harmonics are in phase and therefore produce a significant neutral current, provided there is a path for that current to flow. This effect is most prominent in transformers

depending on their configuration, e.g., non-earthed and delta windings block zero-sequence harmonic currents.

The harmonic power flow can be decoupled as multiple independent power flows for each harmonic $h \in \mathcal{H}$, under the condition (proof: [17] § 16.6) that:

- the overall active/reactive power is the sum of the individual harmonic active/reactive power; and
- the circuit laws, i.e., Kirchhoff's current law and Ohm's law, can be decoupled.

For the purpose of harmonic power flow, it is most natural to use the rectangular current-voltage formulation [18]. The power system of interest is represented by an extended graph $(\mathcal{I}, \mathcal{E}, \mathcal{U})_h$ for each harmonic $h \in \mathcal{H}$.

In what follows, the components of such an extended graph are enumerated and the associated variables, and parameters are introduced, and lastly, the real-valued feasible set for the harmonic power flow is presented.

1) *Buses* $i \in \mathcal{I}$: are the vertices of the graph. The bus voltage is,

$$\bar{U}_{i,h} = U_{i,h}^{\text{re}} + j U_{i,h}^{\text{im}}. \quad (1)$$

The shunt admittance at a bus i is given by $\bar{y}_{i,h}^{\text{sh}}$, where $g_{i,h}^{\text{sh}}$ and $b_{i,h}^{\text{sh}}$ denote the corresponding conductance and susceptance, respectively.

2) *Edges* $e \in \mathcal{E} = \mathcal{B} \cup \mathcal{X}$: are the lines of the graph, consisting of subsets branches $b \in \mathcal{B}$ and two-winding transformers $x \in \mathcal{X}$. A tuple $eij \in \mathcal{T}^{e\rightarrow}$ links an edge e to its from-bus i and to-bus j ; $\mathcal{T}^{e\leftarrow}$ contains the equivalent tuples with i and j reversed, and the union set is $\mathcal{T}^e = \mathcal{T}^{e\rightarrow} \cup \mathcal{T}^{e\leftarrow}$.

3) *Branches* $b \in \mathcal{B} \subset \mathcal{E}$: are represented by a π -model (Fig. 1a). The current flowing through a branch b from bus i to j is,

$$\bar{I}_{bij,h} = I_{bij,h}^{\text{re}} + j I_{bij,h}^{\text{im}} \quad (2)$$

$$= \bar{I}_{bij,h}^{\text{sh}} + \bar{I}_{bij,h}^{\text{s}} \quad (3)$$

$$= (I_{bij,h}^{\text{sh, re}} + I_{bij,h}^{\text{s, re}}) + j (I_{bij,h}^{\text{sh, im}} + I_{bij,h}^{\text{s, im}}). \quad (4)$$

Note that this notation implies $\bar{I}_{bij,h}^{\text{s}} + \bar{I}_{bji,h}^{\text{s}} = 0$. The series impedance of a branch b is given by $\bar{z}_{b,h}^{\text{s}}$, where $r_{b,h}^{\text{s}}$ and $x_{b,h}^{\text{s}}$ denote the corresponding series resistance and reactance, respectively. The shunt admittance of a branch b connected at bus i is given by $\bar{y}_{bij,h}^{\text{sh}}$, where $g_{bij,h}^{\text{sh}}$ and $b_{bij,h}^{\text{sh}}$ denote the corresponding shunt conductance and susceptance, respectively.

4) *Two-Winding Transformers* $x \in \mathcal{X} \subset \mathcal{E}$: are represented by a t -model (Fig. 1b, 1c and 1d) and are split into two parts: core and windings. The current flowing through the core of transformer x from bus i to j is,

$$\bar{I}_{xij,h} = I_{xij,h}^{\text{s, re}} + j I_{xij,h}^{\text{s, im}}. \quad (5)$$

The excitation voltage of transformer x is,

$$\bar{E}_{x,h} = E_{x,h}^{\text{re}} + j E_{x,h}^{\text{im}}. \quad (6)$$

The core captures the effects of

- 1) the vector group through the transformation $t_{x,h}^{\text{vg}}$,
- 2) the series reactance $x_{x,h}^{\text{s}}$, representing the parallel circuit of the leakage inductance L_x^{σ} and stray inter-winding capacitance C_x^{12} , and

³Available at: <https://github.com/timmyfaraday/HarmonicPowerModels.jl>

⁴Note that, for the purpose of this paper, this implies that a potential dc component is omitted.

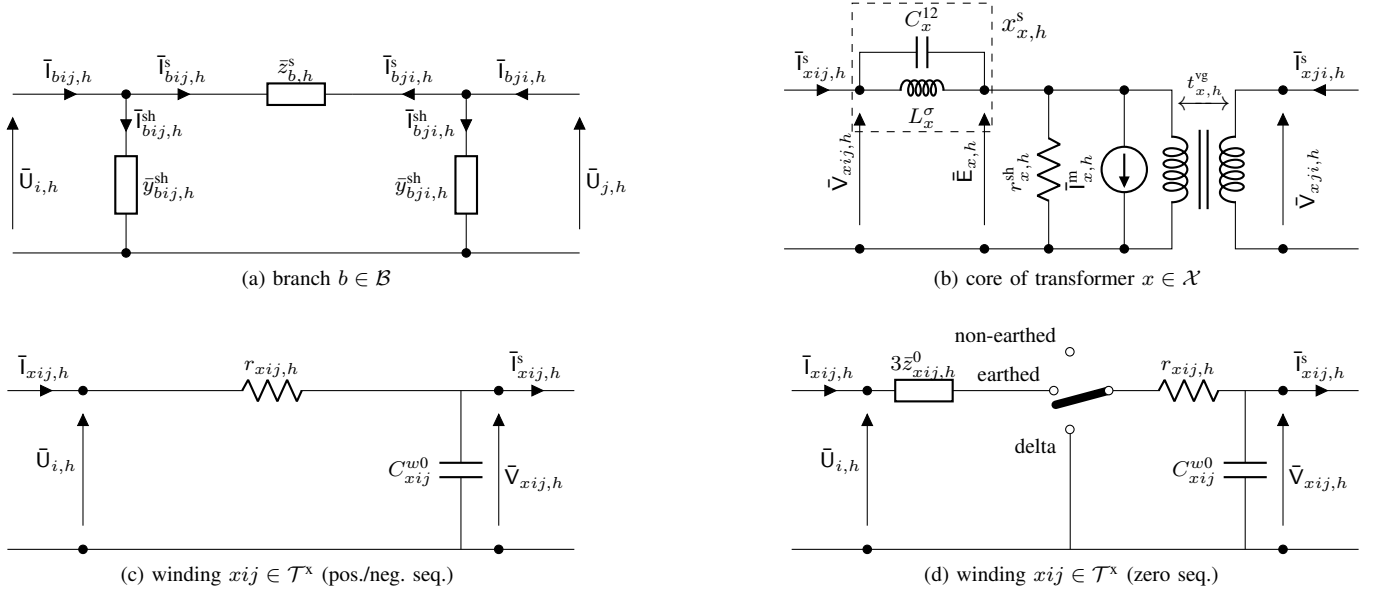


Fig. 1. Equivalent single-phase circuit diagram of (Fig. 1a) branch $b \in \mathcal{B}$ for a given harmonic $h \in \mathcal{H}$, (Fig. 1b) core of a two-winding three-phase transformer $x \in \mathcal{X}$ for a given harmonic $h \in \mathcal{H}$, (Fig. 1c) winding $xij \in \mathcal{T}^x$ of transformer $x \in \mathcal{X}$ for a given positive or negative sequence harmonic $h \in \mathcal{H}^+ \cup \mathcal{H}^-$, and (Fig. 1d) winding $xij \in \mathcal{T}^x$ of transformer $x \in \mathcal{X}$ for a given zero sequence harmonic $h \in \mathcal{H}^0$.

3) the excitation current, consisting of:

- a) the core loss current, modeled through the excitation voltage relative to a shunt resistance $r_{x,h}^{sh}$, and
- b) the magnetizing current $\bar{I}_{x,h}^m = \bar{I}_{x,h}^{m, re} + j \bar{I}_{x,h}^{m, im}$.

Note that, for the purpose of this paper, the magnetizing current is omitted. However, a detailed description on how to include the magnetizing current is available in [16].

The current flowing through a winding $xij \in \mathcal{T}^x$ from node i to the transformer core is,

$$\bar{I}_{xi,j,h} = \bar{I}_{xi,j,h}^{re} + j \bar{I}_{xi,j,h}^{im}. \quad (7)$$

The internal winding voltage is,

$$\bar{V}_{xij,h} = \bar{V}_{xij,h}^{re} + j \bar{V}_{xij,h}^{im}. \quad (8)$$

The windings capture the effects of

- 1) the winding resistance $r_{xij,h}$,
- 2) the stray capacitance between winding and transformer tank C_{xij}^{w0} , and
- 3) the impact of the winding configuration for zero sequence harmonics $h \in \mathcal{H}^0$ (Fig. 1d), with the zero sequence neutral impedance given by $\bar{z}_{xij,h}^0$, where $r_{xij,h}^0$ and $x_{xij,h}^0$ denote the zero-sequence neutral resistance and reactance, respectively.

5) *Units $u \in \mathcal{U}$* : generalize loads and generators. A tuple $ui \in \mathcal{T}^u$ links a unit u to a specific bus i . The current flowing from the bus i to the unit u is,

$$\bar{I}_{u,h} = \bar{I}_{u,h}^{re} + j \bar{I}_{u,h}^{im}. \quad (9)$$

6) *Feasible Set*: In the rectangular current-voltage formulation, the set of linear equality constraints (10)-(31) uniquely

describing the real-valued feasible set for the harmonic power flow of a phase-balanced three-phase power system is,

BUS CONSTRAINTS

reference bus voltage – $\forall i \in \mathcal{I}^{ref}, h \in \mathcal{H}$:

$$\bar{U}_{i,h}^{re} = \bar{U}_{i,h}^{ref, re}, \quad (10)$$

$$\bar{U}_{i,h}^{im} = \bar{U}_{i,h}^{ref, im}, \quad (11)$$

Kirchhoff's current law – $\forall i \in \mathcal{I}, h \in \mathcal{H}$:

$$\sum_{eij \in \mathcal{T}^e} \bar{I}_{eij,h}^{re} + \sum_{ui \in \mathcal{T}^u} \bar{I}_{ui,h}^{re} + \bar{g}_{i,h}^{sh} \bar{U}_{i,h}^{re} - \bar{b}_{i,h}^{sh} \bar{U}_{i,h}^{im} = 0, \quad (12)$$

$$\sum_{eij \in \mathcal{T}^e} \bar{I}_{eij,h}^{im} + \sum_{ui \in \mathcal{T}^u} \bar{I}_{ui,h}^{im} + \bar{g}_{i,h}^{sh} \bar{U}_{i,h}^{im} + \bar{b}_{i,h}^{sh} \bar{U}_{i,h}^{re} = 0, \quad (13)$$

BRANCH CONSTRAINTS

Ohm's law – $\forall bij \in \mathcal{T}^{b \rightarrow}, h \in \mathcal{H}$:

$$\bar{U}_{j,h}^{re} = \bar{U}_{i,h}^{re} - r_{b,h}^s \bar{I}_{bij,h}^{s, re} + x_{b,h}^s \bar{I}_{bij,h}^{s, im}, \quad (14)$$

$$\bar{U}_{j,h}^{im} = \bar{U}_{i,h}^{im} - r_{b,h}^s \bar{I}_{bij,h}^{s, im} - x_{b,h}^s \bar{I}_{bij,h}^{s, re}, \quad (15)$$

branch total current – $\forall bij \in \mathcal{T}^b, h \in \mathcal{H}$:

$$\bar{I}_{bij,h}^{re} = \bar{g}_{bij,h}^{sh} \bar{U}_{i,h}^{re} - \bar{b}_{bij,h}^{sh} \bar{U}_{i,h}^{im} + \bar{I}_{bij,h}^{s, re}, \quad (16)$$

$$\bar{I}_{bij,h}^{im} = \bar{g}_{bij,h}^{sh} \bar{U}_{i,h}^{im} + \bar{b}_{bij,h}^{sh} \bar{U}_{i,h}^{re} + \bar{I}_{bij,h}^{s, im}, \quad (17)$$

TRANSFORMER CORE CONSTRAINTS

Ohm's law – $\forall xij \in \mathcal{T}^{x \rightarrow}, h \in \mathcal{H}$:

$$\bar{V}_{xij,h}^{re} = \bar{E}_{x,h}^{re} - x_{x,h}^s \bar{I}_{xij,h}^{s, im}, \quad (18)$$

$$\bar{V}_{xij,h}^{im} = \bar{E}_{x,h}^{im} + x_{x,h}^s \bar{I}_{xij,h}^{s, re}, \quad (19)$$

Kirchhoff's current law – $\forall xij \in \mathcal{T}^{x \rightarrow}, h \in \mathcal{H}$:

$$\bar{I}_{xji,h}^{s, re} + t_{x,h}^{vg, re} \left(\bar{I}_{xij,h}^{s, re} - \bar{I}_{x,h}^{m, re} - \frac{\bar{E}_{x,h}^{re}}{r_{x,h}^{sh}} \right) + t_{x,h}^{vg, im} \left(\bar{I}_{xij,h}^{s, im} - \bar{I}_{x,h}^{m, im} - \frac{\bar{E}_{x,h}^{im}}{r_{x,h}^{sh}} \right) = 0 \quad (20)$$

$$\begin{aligned} I_{xji,h}^{s,im} + t_{x,h}^{vg,re} \left(I_{xji,h}^{s,im} - I_{x,h}^{m,im} - \frac{E_{x,h}^{im}}{r_{x,h}^{sh}} \right) \\ + t_{x,h}^{vg,im} \left(I_{xji,h}^{s,re} - I_{x,h}^{m,re} - \frac{E_{x,h}^{re}}{r_{x,h}^{sh}} \right) = 0 \end{aligned} \quad (21)$$

transformer phase shift – $\forall xji \in \mathcal{T}^{x\leftarrow}, h \in \mathcal{H}$:

$$E_{x,h}^{re} = t_{x,h}^{vg,re} V_{xji,h}^{re} - t_{x,h}^{vg,im} V_{xji,h}^{im}, \quad (22)$$

$$E_{x,h}^{im} = t_{x,h}^{vg,re} V_{xji,h}^{im} + t_{x,h}^{vg,im} V_{xji,h}^{re}, \quad (23)$$

TRANSFORMER WINDING CONSTRAINTS

Kirchhoff's current law – $\forall xji \in \mathcal{T}^x, h \in \mathcal{H}$:

$$I_{xji,h}^{s,re} = I_{xji,h}^{re} + g_{xji,h}^{sh} V_{xji,h}^{re} - b_{xji,h}^{sh} V_{xji,h}^{im}, \quad (24)$$

$$I_{xji,h}^{s,im} = I_{xji,h}^{im} + g_{xji,h}^{sh} V_{xji,h}^{im} + b_{xji,h}^{sh} V_{xji,h}^{re}, \quad (25)$$

Note that, for (24) and (25), the parameters $g_{xji,h}^{sh}$ and $b_{xji,h}^{sh}$ represent the stray capacitance C_{xji}^{w0} . Alternatively, in case of a delta winding for zero-sequence harmonic $h \in \mathcal{H}^0$, $g_{xji,h}^{sh}$ and $b_{xji,h}^{sh}$ represent the parallel circuit of winding resistance $r_{xji,h}$ and stray capacitance C_{xji}^{w0} (Fig. 1d).

Ohm's law (pos./neg. seq.) – $\forall xji \in \mathcal{T}^x, h \in \mathcal{H}^+ \cup \mathcal{H}^-$:

$$U_{i,h}^{re} = V_{xji,h}^{re} + r_{xji,h} I_{xji,h}^{re}, \quad (26)$$

$$U_{i,h}^{im} = V_{xji,h}^{im} + r_{xji,h} I_{xji,h}^{im}, \quad (27)$$

Ohm's law (zero seq.) – $\forall xji \in \mathcal{T}^{x,earthed}, h \in \mathcal{H}^0$:

$$U_{i,h}^{re} = V_{xji,h}^{re} + (r_{xji,h} + 3r_{xji,h}^0) I_{xji,h}^{re} - 3x_{xji,h}^0 I_{xji,h}^{im}, \quad (28)$$

$$U_{i,h}^{im} = V_{xji,h}^{im} + (r_{xji,h} + 3r_{xji,h}^0) I_{xji,h}^{im} + 3x_{xji,h}^0 I_{xji,h}^{re}, \quad (29)$$

zero seq. current blocking – $\forall xji \in \mathcal{T}^{x,ne} \cup \mathcal{T}^{x,delta}, h \in \mathcal{H}^0$:

$$I_{xji,h}^{re} = 0, \quad (30)$$

$$I_{xji,h}^{im} = 0. \quad (31)$$

B. Harmonic Hosting Capacity

Extending on the harmonic power flow (10)-(31), the harmonic hosting capacity problem determines the maximum amount of harmonic current injected by units at specified buses, given a fairness and harmonic current co-ordination principle, without violating voltage and current bounds as defined in relevant power quality standards.

Two formulations of the harmonic hosting capacity problem are presented (Table II): non-linear and second-order cone, each requiring slightly different variants of certain constraints, as such annotated with *nl* and *soc*, respectively. The second-order cone formulation of the harmonic hosting capacity problem does not consider the fundamental harmonic, rather optionally takes the fundamental power flow as input.

TABLE II

OBJECTIVE, HARMONIC SET, AND CONSTRAINTS OF THE TWO FORMULATIONS OF THE HARMONIC HOSTING CAPACITY PROBLEM.

	non-linear	second-order cone
harmonic set	$\mathcal{H} \subset \mathbb{N}$	$\mathcal{H} \subset \mathbb{N} \setminus \{1\}$
objective	(32) or (33) or (35) or (37)	
fairness	- or (34) or (36) or (38)-(39)	
power flow		(10)-(31)
harm. units	(40)-(43)	(40)-(41)
voltage lim.	(44)-(46)	(47)-(49)
current lim.	(50)	(51)

In what follows, the extensions on the harmonic power flow problem toward the harmonic hosting capacity problem are presented.

1) *Fairness Objective*: The objective of the harmonic hosting capacity problem is to determine the maximum amount of harmonic current that can be injected by units at specified buses. However, given that the harmonic hosting capacity of a power system is limited, the question becomes how to distribute harmonic current injection limits among the connected units in socially justified and economic ways.

The non-linear nature of the Thévenin-equivalent harmonic impedance at any given node introduces a bias between different nodes in terms of 'natural' harmonic hosting capacity. Given the competing interests of the connected users, i.e., the monetary cost of different equipment, and the regulated nature of the electric power industry, more equitable distribution of harmonic hosting capacity seem prudent. This naturally leads to the question of how to (best) balance these interests, which is approached mathematically in a variety of ways, balancing efficiency, equity, fairness and equality [19]. The objective and constraints introduced by a number of fairness criteria are,

$$\text{max. efficiency: } \max \sum_{u \in \mathcal{U}^h} \sum_{h \in \mathcal{H} \setminus \{1\}} I_{u,h}^{\text{mag}}, \quad (32)$$

$$\text{abs. equality: } \max \sum_{u \in \mathcal{U}^h} \sum_{h \in \mathcal{H} \setminus \{1\}} I_{u,h}^{\text{mag}}, \quad (33)$$

$$\text{s.t. } I_{u,h}^{\text{mag}} = I_{v,h}^{\text{mag}}, \forall u, v \in \mathcal{U}^h, h \in \mathcal{H} \setminus \{1\}, \quad (34)$$

$$\text{maximin: } \max \sum_{h \in \mathcal{H} \setminus \{1\}} I_h^{\text{mag}}, \quad (35)$$

$$\text{s.t. } I_h^{\text{mag}} \leq I_{u,h}^{\text{mag}}, \forall u \in \mathcal{U}^h, h \in \mathcal{H} \setminus \{1\}, \quad (36)$$

$$\text{KS bargaining: } \max \sum_{h \in \mathcal{H} \setminus \{1\}} f_h, \quad (37)$$

$$\text{s.t. } I_{u,h}^{\text{mag}} = f_h \hat{I}_{u,h}^{\text{mag}}, \forall u \in \mathcal{U}^h, h \in \mathcal{H} \setminus \{1\}, \quad (38)$$

$$0 \leq f_h \leq 1, \quad \forall h \in \mathcal{H} \setminus \{1\}, \quad (39)$$

where $I_{u,h}^{\text{mag}}$ denotes the harmonic current magnitude of a harmonic unit $u \in \mathcal{U}^h$.

On the one extreme, the maximum efficiency criterion (32) maximizes the injected harmonic current $I_{u,h}^{\text{mag}}$ over all harmonic units $u \in \mathcal{U}^h$ and harmonics $h \in \mathcal{H}$ without taking into account fairness, equality, or equity. On the other extreme, the absolute equality criterion (33) - (34) ensures that the harmonic current $I_{u,h}^{\text{mag}}$ injected by all harmonic units u are equal for all individual harmonics h , likely reducing the efficiency. Balancing efficiency and equality, the maximin criterion (35) - (36) maximizes the minimum harmonic current I_h^{mag} injected by any harmonic units, for all individual harmonics h . Alternatively, the Kalai-Smorodinsky bargaining criterion (37) - (39) maximizes the minimum relative concession by maximizing equal fraction f_h of each harmonic current injected by a harmonic unit relative to its individual maximum harmonic current injection $\hat{I}_{u,h}^{\text{mag}}$, for all individual harmonics h .

2) *Harmonic Unit Model*: In reality, the actual harmonic current at any point on a system is the result of the vector sum of the individual components, originating from harmonic units $u \in \mathcal{U}^h$. Given no a priori knowledge of connected

harmonic units or their harmonic behavior, the coordination of conducted disturbances requires the adaptation of hypotheses relevant to the summation of the disturbances produced by various harmonic units. The lower bound, i.e., minimum harmonic hosting capacity, occurs when all harmonic current injection $\bar{I}_{u,h}$ is in phase relative to a given reference angle $\phi_{u,h}^{\text{ref}}$, accounting for the transformer phase shift [7], and is denoted as,

harmonic current injection – $\forall u \in \mathcal{U}, h \in \mathcal{H} \setminus \{1\}$:

$$\cos(\phi_{u,h}^{\text{ref}}) I_{u,h}^{\text{mag}} = I_{u,h}^{\text{re}}, \quad (40)$$

$$\sin(\phi_{u,h}^{\text{ref}}) I_{u,h}^{\text{mag}} = I_{u,h}^{\text{im}}. \quad (41)$$

Lastly, for the non-linear formulation of the harmonic hosting capacity problem, the fundamental active and reactive power set-point or bounds, i.e., for a load and generator, respectively, are,

nl - fundamental power limits – $\forall ui \in \mathcal{T}^u$:

$$P_u^{\min} \leq U_{i,1}^{\text{re}} I_{u,1}^{\text{re}} + U_{i,1}^{\text{im}} I_{u,1}^{\text{im}} \leq P_u^{\max}, \quad (42)$$

$$Q_u^{\min} \leq U_{i,1}^{\text{re}} I_{u,1}^{\text{im}} - U_{i,1}^{\text{im}} I_{u,1}^{\text{re}} \leq Q_u^{\max}, \quad (43)$$

where $P_i^{\min}$, $P_i^{\max}$, $Q_i^{\min}$ and $Q_i^{\max}$ denote the minimum and maximum fundamental active and reactive power, respectively. A fixed set-point is enforced by setting both the lower and upper bound equal to the intended set-point.

3) *Bus Voltage Bounds*: Relevant standards, e.g., IEC61000-2-4:2002 [20], specify limits for a.o. individual harmonic voltage distortion, total harmonic voltage distortion, and root-mean-square voltage. Depending on the chosen formulation, these limits are enforced as,

nl - root mean square voltage – $\forall i \in \mathcal{I}$:

$$(U_i^{\min})^2 \leq \sum_{h \in \mathcal{H}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_i^{\max})^2, \quad (44)$$

nl - total harmonic voltage – $\forall i \in \mathcal{I}$:

$$\sum_{h \in \mathcal{H} \setminus \{1\}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_i^{\text{thd}})^2 \left((U_{i,1}^{\text{re}})^2 + (U_{i,1}^{\text{im}})^2 \right), \quad (45)$$

nl - individual harmonic voltage – $\forall i \in \mathcal{I}, h \in \mathcal{H} \setminus \{1\}$:

$$(U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_{i,h}^{\text{thd}})^2 \left((U_{i,1}^{\text{re}})^2 + (U_{i,1}^{\text{im}})^2 \right), \quad (46)$$

where $U_i^{\min}$ and $U_i^{\max}$, U_i^{thd} , and $U_{i,h}^{\text{thd}}$ denote for a bus $i \in \mathcal{I}$ the minimum and maximum root-mean-square voltage, total harmonic voltage distortion limit, and individual harmonic voltage distortion limit for harmonic h , respectively, or as,

soc - root mean square voltage – $\forall i \in \mathcal{I}$:

$$\sum_{h \in \mathcal{H}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_i^{\max})^2 - (U_i^{\text{mag,fund}})^2, \quad (47)$$

soc - total harmonic voltage – $\forall i \in \mathcal{I}$:

$$\sum_{h \in \mathcal{H}} (U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_i^{\text{thd}})^2 (U_i^{\text{mag,fund}})^2, \quad (48)$$

soc - individual harmonic voltage – $\forall i \in \mathcal{I}, h \in \mathcal{H}$:

$$(U_{i,h}^{\text{re}})^2 + (U_{i,h}^{\text{im}})^2 \leq (U_{i,h}^{\text{thd}})^2 (U_i^{\text{mag,fund}})^2, \quad (49)$$

where $U_i^{\text{mag,fund}}$ denotes the fundamental voltage magnitude at a bus $i \in \mathcal{I}$, obtained from a preceding fundamental power flow or freely chosen.

4) *Edge Current Bounds*: Relevant standards, e.g., IEC60076-1:2011 [21], specify limits for a.o. root-mean-square current through the edges. Depending on the chosen formulation, these are enforced as,

nl - root mean square current – $\forall eij \in \mathcal{T}^e$:

$$\sum_{h \in \mathcal{H}} (I_{eij,h}^{\text{re}})^2 + (I_{eij,h}^{\text{im}})^2 \leq (I_{eij}^{\text{rated}})^2, \quad (50)$$

where I_{eij}^{rated} denotes the rated current limit of an edge eij , or as,

soc - root mean square current – $\forall eij \in \mathcal{T}^e$:

$$\sum_{h \in \mathcal{H}} (I_{eij,h}^{\text{re}})^2 + (I_{eij,h}^{\text{im}})^2 \leq (I_{eij}^{\text{rated}})^2 - (I_{eij}^{\text{mag,fund}})^2, \quad (51)$$

where $I_{eij}^{\text{mag,fund}}$ denotes the fundamental current magnitude through an edge $eij \in \mathcal{T}^e$, obtained from a preceding power flow or freely chosen.

III. NUMERICAL ILLUSTRATION

This section presents the numerical illustration of the harmonic hosting capacity problem of a phase-balanced three-phase power system. To this end, first, a five bus section of a real-world industrial power system is considered to show the equivalence between the non-linear and second-order cone formulations. Second, a large-scale transmission system is considered, to show the scalability of the proposed second-order cone formulation. For the purpose of these numerical illustrations, any resistive impedance is scaled with a factor $\sqrt{h}$, and any inductive or capacitive impedance is appropriately scaled with a factor h . All results are presented in per-unit using a 100 MVA basis.

A. Industrial Power System

This numerical illustration considers a section of a real-world industrial phase-balanced three-phase power system across five voltage levels, with four transformers, three cables, and four harmonic units (Fig. 2). All necessary data for the harmonic hosting capacity problem can be derived from Fig. 2. On top of the fundamental harmonic, the considered harmonic set is $\mathcal{H} \in \{3, 5, 7\}$, i.e., one harmonic from each sequence subset. The aim of this numerical illustration is to 1) validate the proposed formulations; and 2) show the equivalence between the non-linear and second-order cone models. For the purpose of proving equivalency, both models are solved using IPOPT v1.6.2 [22], where the second-order constraints are translated to their equivalent non-convex quadratic form using the appropriate MATHOPTINTERFACE constraint bridge [23].

1) *Validate the proposed formulation*: It should be noted that the purpose of this numerical illustration is not to provide realistic results for an industrial power system, rather validate correct implementation of the presented mathematical framework. Consequently, for the purpose of validating the proposed formulation, the non-linear model is considered with the Kalai-Smorodinsky bargaining fairness objective. The Kalai-Smorodinsky bargaining fairness objective ensures that (i) individual harmonic voltage distortion at all busses

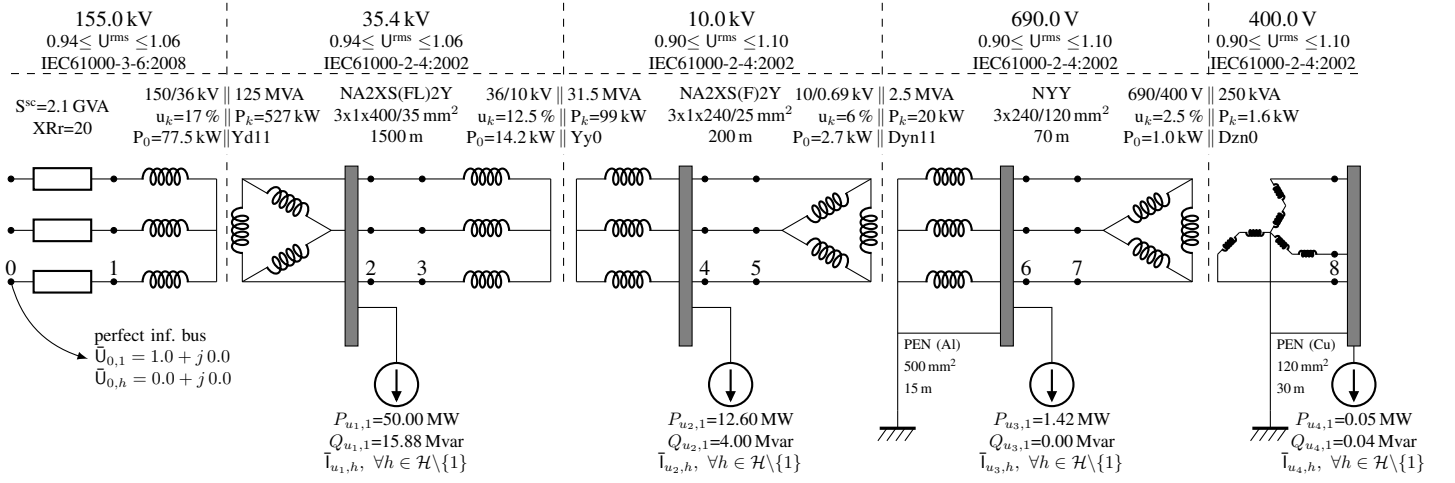


Fig. 2. Section of a real-world industrial phase-balanced three-phase power system across five voltage levels with their root-mean-square voltage limits and applicable harmonic voltage standard, with four transformers, three cables, and four harmonic units with their fundamental power consumption.

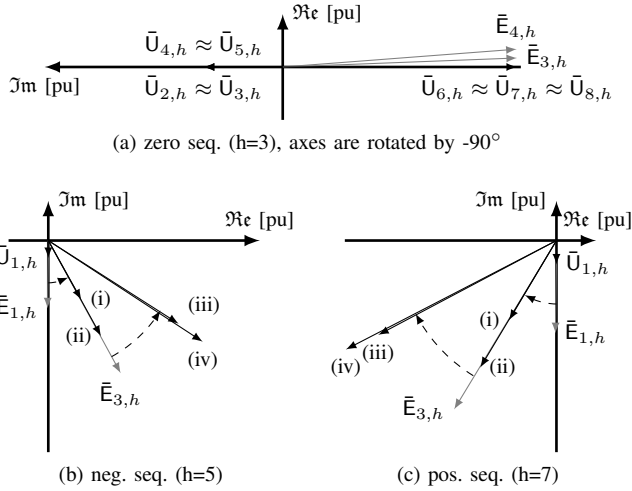


Fig. 3. The bus and excitation voltage phasors throughout the industrial power system (Fig. 2) for (a) zero, (b) negative, and (c) positive sequence, respectively, for the non-linear model with the Kalai-Smorodinsky bargaining fairness objective, where (i) $\bar{U}_{2,h} \approx \bar{U}_{3,h}$, (ii) $\bar{E}_{2,h} \approx \bar{U}_{4,h} \approx \bar{U}_{5,h}$, (iii) $\bar{U}_{6,h} \approx \bar{U}_{7,h}$, and (iv) $\bar{E}_{4,h} \approx \bar{U}_{8,h}$.

approach their limits, resulting in harmonic voltages of similar magnitudes at every bus; and (ii) all units inject at least a minimum amount of harmonic current (Table III), allowing easier validation.

Fig. 3 shows the voltage phasors throughout the industrial power system for the (a) zero (3^{rd} harmonic), (b) negative (5^{th}

harmonic), and (c) positive sequence 37^{th} harmonic, respectively. Fig. 3a shows that, for the zero-sequence harmonics, the transformers successfully decouple the different voltage levels of the system. Fig. 3b and 3c show that the fifth and seventh harmonic voltage $\bar{U}_{1,h}$ at bus one is approximately 25.9 % of the excitation voltage $\bar{E}_{1,h}$ at transformer one, which is inline with the ratio between the short-circuit impedance of the infinite grid and the first transformer. Using a dashed arc, Fig. 3b and 3c show the phase-shift of approximately thirty degrees, respectively counter-clockwise and clockwise, between the excitation and secondary voltage of the first and fourth transformer. Furthermore, a phase-shift of approximately zero degrees between the excitation and secondary voltage of the second and third transformer. Note that the phase-shift is not exactly zero or thirty degrees as it also includes the voltage drop over the secondary winding resistance.

2) *Equivalence between the non-linear and second-order cone model*: Table IV shows the objective value and solve times of the non-linear and second-order cone models for each considered fairness objective. Two take-aways are highlighted: i) the objective value of the non-linear and second-order cone

TABLE III
HARMONIC CURRENT INJECTION FOR THE NON-LINEAR MODEL WITH THE KALAI-SMORODINSKY BARGAINING FAIRNESS OBJECTIVE.

unit	h=3		h=5		h=7	
	$ \bar{I} $ [pu]	$\angle \bar{I}$ [°]	$ \bar{I} $ [pu]	$\angle \bar{I}$ [°]	$ \bar{I} $ [pu]	$\angle \bar{I}$ [°]
u_1	0.0000740	0.0	0.0100655	30.0	0.0098060	-30.0
u_2	0.0000013	0.0	0.0031490	30.0	0.0030833	-30.0
u_3	0.0063852	0.0	0.0005429	60.0	0.0004568	-60.0
u_4	0.0015689	0.0	0.0001167	60.0	0.0000987	-60.0

TABLE IV
OBJECTIVE VALUE AND SOLVE TIMES OF NON-LINEAR AND SECOND-ORDER CONE MODELS, FOR EACH CONSIDERED FAIRNESS OBJECTIVE.

obj.	mod.	obj. value	solve time [s] ⁵			
			pre.	fpf	hhc	tot.
abs. eq.	nl	0.003956251339	-	-	0.53	0.53
	soc	0.003956251338	-	0.06	0.17	0.23
max. eff.	nl	0.097045032196	-	-	0.82	0.82
	soc	0.097045032188	-	0.07	0.35	0.42
maximin	nl	0.000989063236	-	-	0.80	0.80
	soc	0.000989063155	-	0.06	0.26	0.32
KS barg.	nl	1.344028870304	2.12	-	0.45	2.57
	soc	1.344028861750	2.04	0.05	0.14	2.23

⁵The solve times are split in the individual components for the preceding calculation to determine maximum harmonic current injection of each harmonic unit (pre.), fundamental power flow (fpf), and harmonic hosting capacity (hhc), when applicable. Lastly, the total solve time (tot.) is given.

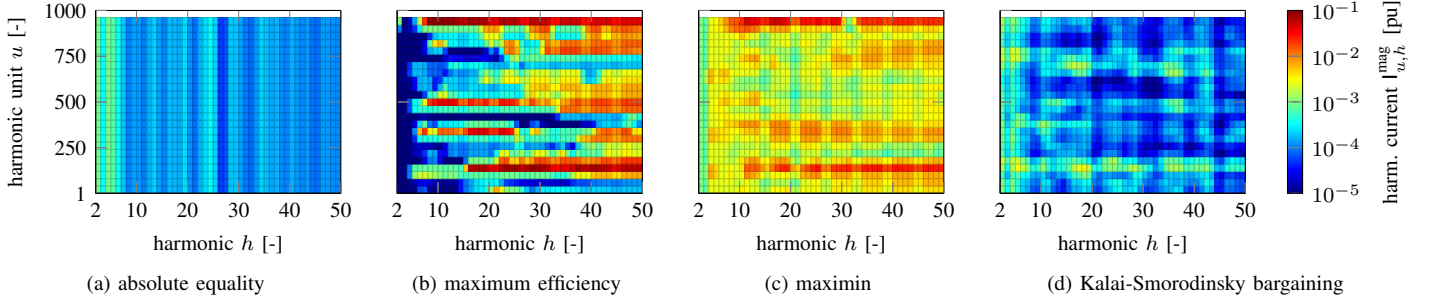


Fig. 4. Results for problem instance (i), including only ihd limits. Heat map of the injected harmonic current $I_{u,h}^{\text{mag}}$ for each harmonic unit $u \in \mathcal{U}^h$ and each harmonic $h \in \mathcal{H}$, for the four considered fairness criteria. Every 40th unit is shown in the figure for the sake of readability.

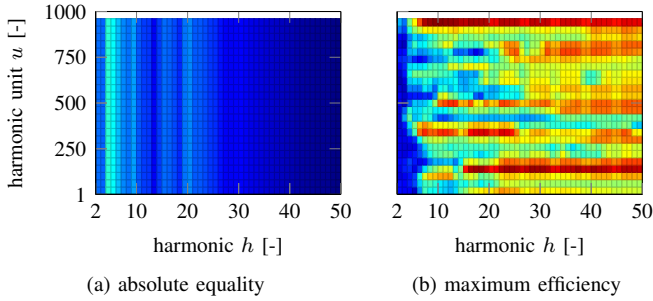


Fig. 5. Results for problem instance (ii), including ihd, thd and rms limits. Heat map of the injected harmonic current $I_{u,h}^{\text{mag}}$ for each harmonic unit $u \in \mathcal{U}^h$ and each harmonic $h \in \mathcal{H}$, for the first two fairness criteria. Every 40th unit is shown in the figure for the sake of readability.

model are equal with an absolute tolerance of at most $1e^{-8}$, proving the equivalence between both models; and ii) even for a test case of limited size and without exploiting a commercial second-order cone solver, it can be seen that the solve times of the second-order cone model are significantly lower than those of non-linear model.

B. Large-scale Transmission System

This numerical illustration aims to demonstrate the developed optimization model on a large transmission grid. The RTE 1888 bus test system, inspired by the French transmission grid is used for that purpose. The test system has twelve voltage levels ranging from 380 kV to 3 kV, 551 transformers, 1980 branches, and 1000 units. All necessary data are derived from the MATPOWER source file [24], and the power quality standard IEC61000-3-6:2008 [10]. As the source file does not specify the transformer vector group, a YNYn0 vector group is assumed for transformers with a primary voltage greater than or equal to 30 kV, while a YNd11 vector group is assumed for all others. The considered harmonic set is $\mathcal{H} \in \{2, 3, \dots, 50\}$. The aim of this numerical illustration is to 1) show the impact of the chosen fairness principle on a power system of realistic size, and 2) show the computational efficiency of the second-order cone model. Only the second-order cone formulation of the harmonic hosting capacity model is considered in this subsection, and is solved using GUROBI v11.0.3.

Two instances of the harmonic hosting capacity model are considered: (i) including only the individual harmonic voltage

distortion (*ihd*) limits, i.e., including (49), and (ii) including individual and total harmonic voltage distortion (*thd*) limits, and root-mean-square (*rms*) voltage and current limits, i.e., including (47)-(49) and (51). Table V shows the a) objective value, b) harmonic hosting capacity lower bound Ξ , and c) computation time, for the harmonic hosting capacity model considering the four proposed fairness criteria (32)-(39). The harmonic hosting capacity lower bound Ξ is determined as,

$$\Xi = \sum_{h \in \mathcal{H} \setminus \{1\}} \sum_{u \in \mathcal{U}^h} I_{u,h}^{\text{mag}}. \quad (52)$$

Table V shows that the absolute equality fairness criterion leads to the lowest harmonic hosting capacity lower bounds because this fairness criterion limits all harmonic currents to have the same value as the lowest allowable harmonic current among them. The Kalai-Smorodinsky bargaining criterion improves the harmonic hosting capacity lower bound by a factor of almost two. However, both the maximin and maximum efficiency fairness criteria lead to significantly higher harmonic hosting capacity lower bounds, as both enable harmonic injections at individual busses to be maximized to the extent allowed by the harmonic voltage limits given by the IEC61000-3-6:2008 standard. Table V also shows that the harmonic hosting capacity lower bounds of the large test system can be calculated in a few minutes for all fairness criteria, demonstrating the computational efficiency of the second order cone formulation of the problem, when the *thd* and *rms* constraints are excluded.

Figure 4 illustrating the injected harmonic current $I_{u,h}^{\text{mag}}$ for a selected number of harmonic units $u \in \mathcal{U}^h$ and each

TABLE V
OBJECTIVE VALUE, HARMONIC HOSTING CAPACITY AND COMPUTATION TIME FOR THE RTE 1888 BUS TEST SYSTEM, CONSIDERING DIFFERENT FAIRNESS CRITERIA.

fairness crit.	(i) w. ihd			(ii) w. ihd, thd, and rms		
	obj.	Ξ [pu]	t [s] ⁶	obj.	Ξ [pu]	t [h]
abs. eq.	13.07	13.07	307.5	3.59	3.59	85.3
max. eff.	630.02	630.02	142.7	575.70	575.70	4.6
maximin	0.04	305.91	101.07	*	*	*
KS barg.	0.31	24.61	131.9	*	*	*

⁶For the KS bargaining approach, the maximum allowable nodal harmonic current injection for an individual unit $\hat{I}_{u,h}^{\text{max}}$ is calculated in about two hours.

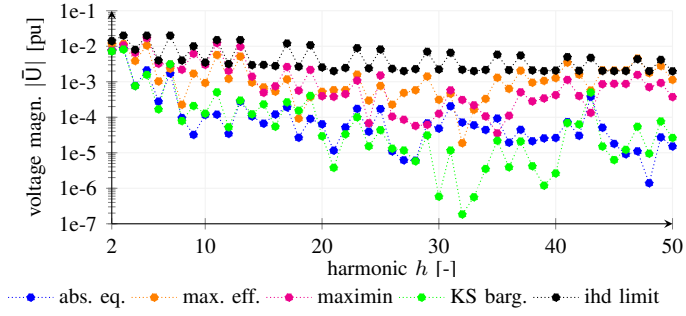


Fig. 6. Nodal harmonic voltage on the 380 kV bus (bus 127) for the different fairness criteria, and ihd limits (IEC61000-3-6:2008).

harmonic $h \in \mathcal{H}$ demonstrates how the fairness criterion affects the distribution of the harmonic budget both throughout the grid and over different harmonics.

Table V further demonstrates the effect of enforcing the *thd* limit and *rms* constraints on the harmonic hosting capacity problem. The absolute equality and maximum efficiency criteria have been used to demonstrate the effect. Expectedly, it is observed that the harmonic hosting capacity lower bound is reduced in comparison to only enforcing individual harmonic voltage distortion limits. Furthermore, the inclusion of *thd* and *rms* constraints increase the computation time drastically, which can be attributed to the fact that multiple solutions with the same *thd* value can exist, without having an influence on the objective function. This leads to a numerical degeneration of the problem resulting in very large computation times. A comparison of Figure 4b and Figure 5b shows that despite a reduction of the harmonic current injections in the instance with *thd* and *rms* constraints, the overall relative distribution throughout the grid and over the harmonics remains similar to the instance without the inclusion of the *thd* and *rms* constraints.

The effect of the chosen fairness criterion is also demonstrated in Figure 6, where the nodal harmonic voltage of a selected 380 kV bus (bus 127) obtained for each fairness criterion using individual harmonic voltage limits is depicted, along with the maximum allowable nodal harmonic voltage calculated using the IEC61000-3-6:2008 standard and the harmonic impedance of the selected node (Fig. 7). We can observe that the maximum efficiency and the maximin criteria succeed in better utilising the harmonic budget without violating limits provided in the standard. Considering that the KS bargaining and the absolute equality criteria are more affected by the grid nodes with lower harmonic budgets they under utilize the available harmonic hosting capacity. The average harmonic voltages of the system show that the maximum efficiency criterion most efficiently utilizes the harmonic budget (Fig. 8).

IV. CONCLUSION

In this paper we introduce a tractable second order cone formulation of the harmonic hosting capacity problem for balanced three-phase networks which uses harmonic current injections in rectangular coordinates for the all harmonics in

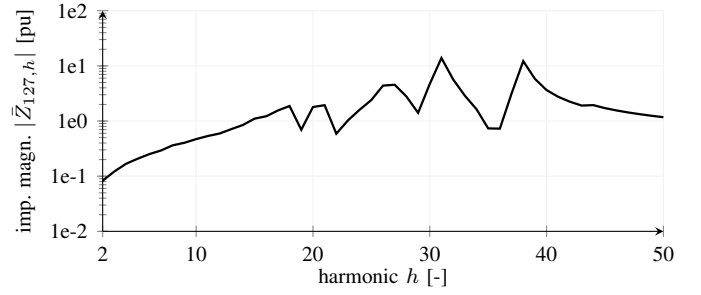


Fig. 7. Nodal harmonic impedance of the selected 380 kV node (bus 127)

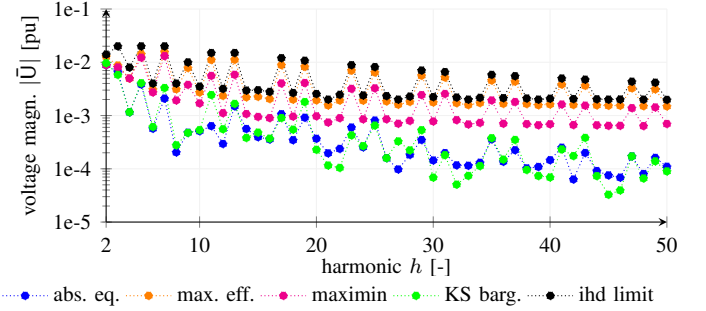


Fig. 8. Average harmonic voltage magnitude for the different harmonics and fairness criteria and ihd limits (IEC61000-3-6:2008).

accordance with the existing power quality standards. The problem formulation allows to determine a lower bound on the harmonic hosting capacity under the assumption of equal phase angles of all harmonic injections, which does not allow for harmonic compensation. It is numerically demonstrated that the second-order cone formulation is equivalent to the nonlinear formulation of the problem using an industrial network as a test case. Furthermore, the numerical results obtained the 1888 bus Rte test system show that this approach can successfully be applied to realistic country-scale networks.

It is further shown that the choice of the fairness criterion for allocating the harmonic budget results in substantially different system harmonic hosting capacities which should be addressed in future editions of power quality standards. The allocation of the harmonic budget will gain evermore importance with the massive expected integration of inverter based resources in the future.

Furthermore, although particular fairness criteria result in substantially different harmonic spectra on the nodal level, the average harmonic voltages on the system level behave similarly despite having different individual magnitudes.

The presented models do not yet capture the stochasticity of injections or uncertainties around grid parameters affecting the harmonic hosting capacity of a power system. In future work, the authors will work towards including both the stochastic nature of the power system itself, e.g., component parameters or power system exploitation, as well as the angle of the harmonic current injected by the units.

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